CIE – II – 2025

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	27th January 2025	Maximum Marks	10 + 50
Course Code	21HS71	Duration	120 Mins
Sem	VII Semester	CIE – II	
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Define Uniform Civil Code.	02	2	2
2	List any two powers of National Human Rights Commission (NHRC) .	02	2	2
3	What is "product liability"?	02	2	2
4	Differentiate between unfair Trade practice & Deficiency in Services.	02	2	2
5	List any two consumer Rights available under the Act	02	2	2

PART – B

Sl. No.	Questions	M	BT	CO
1.a	Examine the Constitutional position of Governor of a State.	05	2	1
1.b	Discuss Online shopping and product liability covering-Liability of a product retailer or delivery service and Liability related to online food products.	05	2	1
2.a	Explain the Impeachment proceedings of President of India.	05	2	2
2.b	Discuss the ground and procedure of removing the Judges of Supreme Court	05	3	2
3	List 11 Fundamental Duties enshrined in the Constitution of India.	10	2	1
4	The slogan line of "Jago Grahak Jago" is applicable today. Discuss the Historical perspective, Significance and Key features of Consumer Protection Act (both old and new).	10	3	2
5	Directive principle is fundamental in making Welfare state". Discuss	10	4	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	--	10	--	--	--	--	10	--	--	--	--
		Test	20	20	10	-	25	15	10		-	-	20

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	27th January 2025	Maximum Marks	10 + 50
Course Code	21IM72	Duration	120 Mins
Sem	VII Semester	CIE – II	

PRODUCT DESIGN & DEVELOPMENT

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	What does Design for Cost (DfC) aim to achieve?	01	L_2	CO3
2	List any two main objectives of Design for Maintainability (DfM)?	01	L_2	CO3
3	Name one method used in Design for Usability (DfU).	01	L_1	CO3
4	Mention the purpose of Design for Sustainability (DfS)?	01	L_2	CO3
5	How does Design for Logistics (DfL) impact the supply chain management?	01	L_2	CO3
6	What is the focus of Design for Compliance (DfC)?	01	L_1	CO4
7	Mention Design for Testability (DfT) improve product quality?	01	L_2	CO4
8	Define "product backlog" in agile.	01	L_1	CO4
9	Which agile role is responsible for maximizing the value of the product?	01	L_1	CO4
10	Mention any one feature of the stage gate product development process.	01	L_2	CO4

PART – B

Sl. No.	Questions	M	BT	CO
1.	Explain the principles of Design for Assembly (DfA). How does DfA impact the efficiency and cost of product assembly? Use case studies or examples to support your answer.	10	L_3	CO3
2.	A company is developing a new eco-friendly packaging solution. Evaluate how Design for Sustainability (DfS) principles can be integrated into their product development process and the potential impact on their brand and market position.	10	L_4	CO4
3.	A consumer electronics company is launching a new smartphone and wants to ensure it is easy to assemble and test. Discuss how Design for Assembly (DfA) and Design for Testability (DfT) principles can be applied to achieve these goals.	10	L_4	CO4
4.	A global automotive company is developing an electric vehicle. Discuss how they can incorporate Design for Cost (DfC) and Design for Sustainability (DfS) principles to ensure the vehicle is affordable and environmentally friendly.	10	L_4	CO4
5.	Discuss the advantages and disadvantages of using the agile product development approach in a fast-paced technology environment. Provide examples to support your arguments.	10	L_3	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	-	-	05	05	--	4	6	-	--	--	--
		Test			20	30	--	--	--	20	30		--

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**50 + 10**
Course Code	**21IM74HD**	**Duration**	**90 + 20 Min**
Sem	**VII**	**CIE – II**	

DATABASE MANAGEMENT SYSTEM

Note: Answer all the Questions.

Sl No	Questions	M	BT	CO
	Part A			
1.	Enumerate the three types of relationship that exists between entities in an ER diagram.	02	L1	CO2
2.	Mention the need for defining Database constraints in DBMS.	02	L4	CO1
3.	List the features of Foreign Key.	02	L2	CO3
4.	PROJECT is Unary operation. Justify	02	L4	CO3
5.	Consider the following schema of relations: *Student (ssn, name, address, major)* *Course (code, title)* *Registered (ssn,code)* Write an appropriate query for listing the codes of courses in which at least one student is registered (registered courses).	02	L3	CO4
	Part B			
1 a	Consider the following schema: Suppliers (sid : integer, sname : string, address :string) Parts (pid : integer, pname : string, color : string) Catalog (sid : integer,pid : integer, cost : real) i. Find the name of suppliers who supply some red parts ii. Find the sids of suppliers who supply every part	05	L3	CO4
b	Explain the integrity constraints: Not Null, Primary Key with an example each. Is the combination 'Not Null, Primary Key' a valid combination? Justify.	05	L4	CO3
2	Illustrate different types of Constraints in DBMS with examples.	10	L3	CO1
3	"Whenever update operations are applied on the relations, it is necessary that the integrity constraints are not violated". Evaluate the statement with relevant examples.	10	L4	CO3
4	Consider the following relation, BOOK.	10	L3	CO4

BOOK	Acc-No	Yr-Pub	Title
	734216	1982	Supply Chain Management
	237235	1995	Database design
	631523	1992	Machine design
	543211	1991	Quality Management
	376112	1992	Personnel Management

Write the syntax for applying the following queries on the relation, BOOK.

a) Select from the relation, BOOK all the books whose year of publication is 1992.

b) Select from the relation, BOOK all the books whose Acc-No is greater than

	or equal to 56782. c) List all the Title and Acc-no of the BOOK relation. List all the Title and Yr-Pub of the BOOK relation			
5	From the relations R and S , find the following: (a) $R \cup S$ (b) $R \cap S$ (c) $R - S$ (d) $S - R$	10	L4	C

R	Sl.No	DEPT
	S1	PHYS1
	S2	PSY
	S3	CHEM
	S4	JOUR

S	Sl.No	DEPT
	S10	MATH1
	S3	CHEM
	S15	ENG
	S16	MATH2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	2	2	4	2	-	02	02	02	04	-	-
	Test		10	-	15	25	-	-	-	25	25	-	-

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	28.01.2025	Maximum Marks	10+50
Course Code	21IM74HA	Duration	30+90 Min
Sem	VII	CIE – II	

FACILITIES PLANNING and DESIGN

Note:

1. Answer all the Questions.

	Part – A			
Q.No	Questions	M	CO	BT
1.1	Define a unit load as per Tanchoco et al.	(02)	L1	CO1
1.2	Small loads _____ the transportation requirements but can potentially _____ work-in-process inventory.	(02)	L2	CO1
1.3	List any two basic methods of moving the unit load according to Apple.	(02)	L1	CO1
1.4	What are the critical types of information needed to integrate a plant-wide material handling system	(02)	L1	CO1
1.5	Write in one sentence the JIT Impact on Facilities Design.	(02)	L1	CO1

	Part –B			
Sl. No.	Questions	M	BT	CO
1	Enumerate and elaborate the steps in material handling system design.	10	L2	CO1
2	Given the following dimensions of a particular type of plastic reusable containers Inside dimensions 18" 11" 11" Outside dimensions 20" 12" 12" Each nested container 20" 12" 2" A trailer with inside dimensions of 240" 120" 120" is used to transport these containers. The containers are not palletized. Assume that no clearance is needed between containers or between containers and the walls of the trailer. The dimension of the storage opening is 24" 16" 14". Determine the following: i. Container space utilization ii. Storage space efficiency iii. Container nesting ratio iv. Trailer space utilization if all containers are stacked vertically in only one orientation v. Trailer return ratio	10	L3	CO3
3	Explain the guidelines for estimating the material handling costs in practice.	10	L2	CO2
4	Summarize the features of fixed automation systems. Evaluate the merits, demerits and applications of such systems.	10	L3	CO3
5	List the manufacturing situations for which a flexible manufacturing system (FMS) might be appropriate. What are the various means of achieving flexibility in FMS? List and illustrate with sketches the alternative layouts for a FMS.	10	L1	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	10				8	2				
	Test		20	10	20		10	20	20			

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM75IM**	**Duration**	**30 + 90 Min**
Sem	**VII**	**CIE – II**	

PROJECT MANAGEMENT (Institutional Elective-II)

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Mention the inputs required to create project charter process.	02	L1	CO1
2	What is work performance data?	02	L2	CO1
3	Give any two examples of prevention costs in cost of quality.	02	L3	CO2
4	Mention the use of affinity diagram in managing the quality.	02	L2	CO1
5	What is secondary risk? Give an example.	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Assume that you are developing a **Student Management Information System** (SAP) for your institution. Create a project charter for the above case. Also discuss what are the issues with the current SAP system.	15	L4	CO3
2.	You are leading a project to develop a new e-commerce platform. During the initial stages, your stakeholders are unclear about the scope and objectives. a) Identify the key document you need to create to formally authorize the project and outline its purpose. b) Explain the process you would follow to ensure all stakeholders are aligned with the project scope and objectives.	10	L3	CO4
3.	Compare and contrast qualitative and quantitative risk analysis techniques used in project risk management, providing examples of when each approach is most appropriate.	10	L3	CO2
4.	Evaluate the potential consequences for a company that neglects to address quality costs in its operations, considering both short-term and long-term implications. Take a case example of any project of your choice.	10	L3	CO3
5.	List seven quality tools used to manage project.	05	L1	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	--	--	--	02	06	02	--	--	--
		Test	--	15	25	10	--	05	--	30	15	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	28th January 2025	Maximum Marks	10 + 50
Course Code	21IM73G1	Duration	120 Mins
Sem	VII Semester	CIE – II	

DATA MINING AND PREDICTIVE ANALYTICS (Elective)

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is Association Rule Mining in data mining?	01	L_2	CO2
2	Which metric measures the strength of the association rule?	01	L_1	CO2
3	Define the term 'support' in the context of Association Rule Mining.	01	L_1	CO2
4	What does the acronym CART stand for in data mining?	01	L_1	CO2
5	What does the term 'confidence' refer to in Association Rule Mining?	01	L_2	CO2
6	What type of variable does a regression tree predict?	01	L_2	CO3
7	What does 'Gini index' measure in decision tree analysis?	01	L_2	CO3
8	In K-Means clustering, 'K' stand for_____	01	L_2	CO3
9	What metric is commonly used to measure distance in K-Means Clustering?	01	L_2	CO3
10	Give one limitation of the K-Means Clustering algorithm.	01	L_2	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Explain the Apriori algorithm used in association rule mining. Discuss its importance and provide a step-by-step example to illustrate its working.	10	L_3	CO2
2.	Given a dataset of transactions: - Transaction 1: {A, B, D} - Transaction 2: {B, C} - Transaction 3: {A, B, D, E} - Transaction 4: {A, C, D} - Transaction 5: {B, D, E} Calculate the support, confidence, and lift for the association rule {B} → {D}. Show all steps and calculations in your solution.	10	L_4	CO2
3.	A retail store wants to predict whether a customer will make a purchase based on their browsing behavior. The store collects the following data:	10	L_4	CO2

Time Spent on Site (minutes)	Number of Pages Visited	Made Purchase (Class Label)
5	3	Yes
2	1	No
8	5	Yes
1	1	No
7	4	Yes

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73GC**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	

VIRTUAL COMMERCE (Departmental Elective)

Note: Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by a protocol in an internet architecture context?	02	L2	CO1
2	What is internet economy?	02	L2	CO1
3	Mention any two features of B2G e-commerce model	02	L1	CO2
5	Uniform resource locator (URL) is used for providing _____ for a web page.	01	L2	CO2
6	Apart from considering factors relating to the firm's competitive environment, it is also necessary to consider factors relating to the firm's _____	01	L2	CO1
7	Social commerce is a concept where electronic commerce is used in conjunction with _____	01	L1	CO2
8	Any business that expects to grow profitably and effectively will have to master the art of selling through a _____ of multiple channels.	01	L1	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	"Among the more evident benefits of e-markets is the increase in price transparency." Justify this statement with suitable case example	10	L3	CO3
2.	Evaluate the three major forces fueling e-commerce?	10	L4	CO4
3.	How can mass customization concept impact the efforts of taking any business to V-commerce platform?	10	L2	CO2
4.	Explain the advantages of presence of V-commerce in multichannel marketing.	10	L2	CO1
5.	Explain the driving Factors of B2C Online-Selling applied to Indian markets, targeting rural customers	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	--	--	--	04	06	--	--	--	--
		Test	10	20	10	10	--	--	30	10	10	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29/01/2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM75IN**	**Duration**	**30 + 90 Min**
Sem	**VII**	**CIE – II**	

SUPPLY CHAIN ANALYTICS (Institutional Elective II)

Note:

1. Answer all the Questions.
2. Use of Statistical table permitted

Part – A

Sl. No.	Questions	M	BT	CO
1	Enumerate any 4 supply chain risks.	02	02	03
2	Identify any 4 important factors to be considered for warehouse layout design.	02	02	03
3	A company operates with an average lead time of 4 days and a daily average demand of 100 units. The calculated safety stock is 476 units. Calculate the Reorder Point (ROP).	02	03	03
4	What are the four main stages of the SPSS model in demand management?	02	02	04
5	State the formula used for calculating MAPE in demand forecasting.	02	02	04

Part –B

Sl. No.	Questions	M	BT	CO
1	Choose a manufacturing firm and a service provider, and compare and contrast which one has a higher degree of vertical integration, and which one has more outsourcing? Discuss the pros and cons of vertical integration and outsourcing with your chosen companies	10	3	2
2	Managing relationships in a multi-tier supply network can be very complex and challenging. Discuss the various relationship types in a network and approaches to managing them.	10	2	3
3	Discuss the role of a Warehouse Management System (WMS) in optimizing warehouse operations. Highlight its key benefits with examples. Additionally, identify and explain five Key Performance Indicators (KPIs) commonly used to measure warehouse performance and their significance in improving operational efficiency.	10	3	3
4	Analyze the various demand forecasting methods by categorizing them into qualitative and quantitative approaches. Examine the key differences between these two types of methods and assess how they impact decision-making in different business scenarios.	10	4	4
5 a	A company operates with an average daily demand D_{avg} of 100 units and an average lead time LT_{avg} of 4 days. The standard deviation of daily demand σ_D is 20 units, and the standard deviation of lead time σ_{LT} is 2 days. The company aims to achieve a 95% service level. Using the given data, calculate the safety stock required to meet the target service level. Z value is 1.96	05	2	3
b.	Holding inventories can be costly. Identify the various costs associated with inventory holding within a company of your choice	05	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	--	--	06	04	--	08	02	--	--	--
	Test		--	10	30	10	--	15	25	10	--	--

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Department of Electronics & Telecommunication Engineering

Date	29th Jan. 2025	Maximum Marks/Quiz	50+10
Course Code/Title	21ET75IK	Duration	120 Min
Sem	VII Semester	CIE-II	

SPACE TECHNOLOGY AND APPLICATIONS

Q.No	PART A	M	BT	CO
1.	Mention the spectrum range and application of Mid Infrared bands in Remote sensing?	1	L2	CO3
2.	What is the difference between supervised and unsupervised image classification.	1	L2	CO3
3.	The 16 bit image consists of---- --number of pixel values.	1	L1	CO3
4.	The function of Handoff is___	1	L2	CO2
5.	Communication satellites and weather satellites are often placed in ____Orbit.	1	L2	CO2
6.	RADAR is ------ type of remote sensing.	1	L2	CO3
7.	What is temporal and Radiometric resolution?	2	L2	CO3
8.	List any two recent Indian satellites that are designed for extra-terrestrial scientific observations.	2	L3	CO4
	PART B			
1.	What are LEO, MEO and GEO orbits? Explain with relevance to applications. Mention at least one example of a satellite launched by India to each of the three orbits.	10	L2	C2
2. a)	Draw the block diagram showing a home terminal for DBS TV/FM reception and explain its principles.	06	L2	C3
b)	The orbit for an earth orbiting satellite has an eccentricity of 0.15 and a semi major axis of 9000 km. Determine: i) The Apogee height. ii) The Perigee height. (Assume R=6370 km.)	04	L3	C2
3.	Describe the functioning of Attitude and Orbit Control (AOCS) system in satellite and also explain different methods are used to control.	10	L3	C2
4.a)	Differentiate four types of multiple access techniques used in satellite transmission.	05	L2	C2
b)	Explain the principles of VSAT and their applications.	05	L2	C2
5.	What is Remote sensing? Explain two types of Remote sensing instruments. List out important advantages and disadvantages of Remote sensing.	10	L2	C3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6

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NBA Accredted (UG - 6 Years)

Department of Electronics & Communication Engineering

Date: 7/2024	Test - 2	Max. Marks:10+50
Semester:VI	UG	Duration: $1\frac{1}{2}$ Hrs
Course: Human Machine Interface(Institutional Elective)		Code: 21IE6F10

S No	Part A	M	BT	CO
1.	UX concentrates on----------	1		
2.	Simple sketches or digital mockups that show the layout, structure, and organization of content on a webpage or app screen are called----------	1		
3.	Name any 2 prototyping tools for UX----------	1		
4.	Fictional characters that represent the target users of a product, used to guide design decisions and keep the user's needs in mind are known as----------	1		
5.	The style and formatting of text used in a product, including font families, sizes, and weights is called----------	1		
6.	Cross-platform application programming interface (API) for rendering 2D and 3D vector graphics is called----------	1		
7.	Which of the following principles is NOT a key component of a user-cantered HMI development process----------	1		
8.	In the context of a user-centered HMI development process, which of the following methods is primarily used to gather detailed requirements directly from users?	1		
9.	Which of the following best describes the principle of 'Feedback' in mobile UI design?	1		
10.	Which of the following is NOT a common function of a web server?	1		

Department of Electronics & Communication Engineering

Date: .07.2024	Test – II	Max.
Semester: VI	UG	Dura
Course: Human Machine Interface(Institutional Elective)		Code:

S No	Questions
1.a	List and explain the 5 elements of UX in detail with neat diagrams.
2. a	Illustrate the five core components of UX designing in detail.
3.	Illustrate the seven phases of Design Thinking process in detail.
4. a	List and briefly discuss the following: i. 5 dimensions of interactive design ii. Any 5 UX design asset types
b	What is a web server? Describe in brief the main components of a web server
5.	List any three types of HMIs. List the main stages of the user-centered HMI development process and explain the importance of each stage

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2024-2025 (Odd Sem)
Common to all branches

Date	January, 2025	Maximum Marks	10 + 50
Course Code	21BT75IB	Duration	120 min
Sem	VII Semester	Closed Book Offline Quiz and Test – 2	

Healthcare Analytics (Institutional Elective)

Answer all questions in Part A & Part B. Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	Multiple sequence analysis consists _____ or more number of sequences.	1	2	1
2	Needleman_Wunsch algorithm can be applied for _____ alignment.	1	2	1
3	Expand BLAST.	1	2	2
4	Among the PAM1 and PAM250, which one has the more distant?	1	3	2
5	Differentiate between the opening and extension gap penalties.	2	2	3
6	What is null hypothesis?	2	4	3
7	Mention any 2 measures of statistical significance.	2	2	4

Part B

Sl. No	Questions (Test)	M	BT-L	CO
1.	What are the different types of sequence alignments? Explain any one in brief.	10	2	3
2	How does the scoring matrix helps to derive the interpretation among sequence alignment?	10	3	3
3	Elucidate on the profiles and Hidden Morkov Models for bioinformatics data.	10	4	4
4	Differentiate between PAM and BLOSSUM and write its significance.	10	4	4
5	Write an overview of molecular phylogeny.	10	2	3

BTL-Blooms taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	~~42~~	2	~~60~~ 36	40 22		~~40~~ 27	~~20~~ 18	~~48~~ 20		

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	29th Dec 2024	Maximum Marks	50
Course Code	21EE75IH	Duration	90 Mins
Sem	VII Semester	TEST-II	
E-Mobility			

Q.No	Part B – Test Questions	Marks	COs	BT
1a.	Discuss the different techniques employed in BMS to measure the following battery parameters: (a) Current and (b) Temperature	06	02	01
1b.	Why battery management systems have to be designed for EV? Explain briefly.	04	01	01
2a.	With a neat circuit diagram explain the process of active cell balancing process.	06	02	03
2b.	Explain briefly about the causes of cell imbalance.	04	02	01
3a.	Analyze in detail about CV chargers.	06	03	03
3b.	What are balancers? Explain its applications.	04	02	04
4a.	Discuss any three issues related to energy management strategies implementation.	06	03	02
4b.	Discuss briefly about the classification of various energy management strategies.	04	01	01
5a.	Explain in detail about deep reinforced learning energy management algorithm.	06	03	03
5b.	Compare the following Energy Management strategies: (a) rule-based strategy and (b) Optimization Based.	04	03	03

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	08	20	22	-	18	06	22	04	-	-

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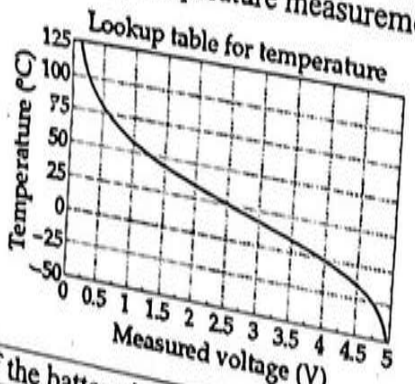
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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	29th Dec 2024	Maximum Marks	
Course Code	21EE75IH	Duration	
Sem	VII Semester		

E-Mobility QUIZ-II

Q.No.	Part A – Quiz Questions	Marks	CO
1.	Define energy management strategy in the context of electric vehicles.	02	2
2.	List the essential functions of a BMS.	02	2
3.	What are the primary inputs for BMS to estimate the SOC.	02	2
4.	Why are energy management strategies critical in hybrid and electric vehicles?	02	3
5.	By considering the below graph identify which type of thermistor is employed for temperature measurement 	01	3
6.	What happens if the battery is allowed for overcharging?	01	3

Marks Distribution	Particulars		BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks								
	Test	Max Marks	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5
			-	06	04	-	06	03	01	-	-

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Department of Civil Engineering

Date	29-01-2025	Maximum Marks	50+10
Course Code	21CV751F	Duration	120 Min
Sem	VII	CIE – II	
INTEGRATED HEALTH MONITORING OF STRUCTURES			

Note: Answer all questions.

QUIZ

Sl. No.	Questions	M	BT	CO
1	Define smart materials in the context of Structural Health Monitoring (SHM)	2	1	2
2	What is Static field testing	2	1	3
3	List two Electric Field Sensors	2	1	2
4	List the Benefits of AI in SHM	2	1	2
5	Load cell can be used as _____, _____	2	1	2

TEST

Sl. No.	Questions	M	BT	CO
1	With a neat sketch explain the evolution of materials used for structural health monitoring and its application .	10	3	2
2	Classify Smart Materials used in SHM with examples	10	3	2
3	Explain the different types of static load tests and conditions for its application	10	2	3
4	Discuss the sensor systems and hardware requirements in Static Field Testing	10	2	3
5	Explain the working Principle and advantage of EMI Technique for structural health monitoring	10	2	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	38	22	-	10	30	20			

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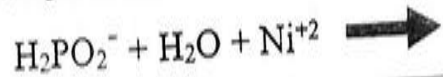
(CIE-II FOR VII SEM)

DEPARTMENT OF CHEMISTRY

Date	29.01.2025	Sem - VII	CIE-II
Course Code	21CM751D	Maximum Test Marks	Quiz-10 + Test-50 (60M)
Course Name	ACSM	Duration	120 mins

ADVANCES IN CORROSION SCIENCE AND MANAGEMENT

Note: 1) Answer all the questions
2) Part A should be answered in the first two pages of the answer script.

	Part-A	M	BTL	CO
1	What is the impact of substrate temperature on the PVD (sputtering) process?	1	4	2
2	How does the alkalinity of an environment influence the interpretation of Pourbaix diagrams?	1	3	2
3	What is the role of resin in varnish?	1	2	1
4	How do you differentiate between varnish and enamel?	1	3	3
5	Chromium anodes are not used in the process of chrome plating. Justify	1	3	2
6	What are oxygen scavengers?	1	2	1
7	Identify the missing product in the following reaction concerning the electroless plating of Nickel. 	1	4	2
8	Write the electrolytic bath composition of Nickel plating.	1	1	2
9	Name the film-forming material in paint.	1	1	2
10	List any one limitation of the Pourbaix diagram.	1	1	3

	Part-B	M	BTL	CO
11	Outline the methods by which corrosion inhibitors function to reduce corrosion rates in metal structures. Explain with reactions.	7	2	2
12	For the oxygen reduction reaction $O_2(g) + 4\,H^+(aq) + 4e^- \longleftrightarrow 2H_2O(l)$ The standard electrode potential (Eo) is +1.23 V. Calculate the equilibrium potential at pH= 7, where the concentration of oxygen is 0.21 atm. Interpret the result obtained.	7	3	3

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Academic year 2024-2025 (Odd Sem)
(CIE-II FOR VII SEM)

13	What are the key differences between electroless plating and electroplating? Explain the process of copper electroplating with reactions.	7	
14	List out the main components of paints. Explain any 3 components with examples along with their role.	7	
15	How can corrosion be controlled through effective design? Analyze and interpret the role of design in corrosion control concerning practical situations. what is the clarity of this question	7	4
16	An experiment involved two aluminum samples, each with an initial weight of 200 grams. One sample was treated with a corrosion inhibitor, while the other remained untreated. After exposure to a marine environment for 30 days, the treated sample's final weight was 198 grams, and the untreated sample's final weight was 195 grams. Given the density of aluminum is 2.7 g/cm³ and the constant K is 87.6, calculate the corrosion rate for each sample using the weight loss method, expressed in millimeters per year (mmpy). Compare and interpret the results to determine the effectiveness of the corrosion inhibitor.	7	3
17	Construct a Pourbaix diagram for Aluminium, assuming the activity of all dissolved substances is 10^{-6} M at room temperature (298K). Using the diagram, interpret the corrosion tendency of Aluminium under different conditions, including the relevant electrochemical reactions. Discuss practical solutions for mitigating Aluminium corrosion based on your diagram analysis.	8	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

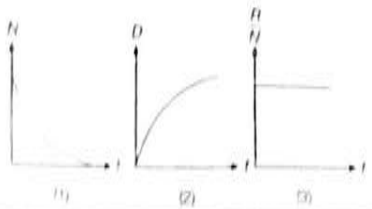
Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Max Marks	16	13	16	15	4	16	25	15

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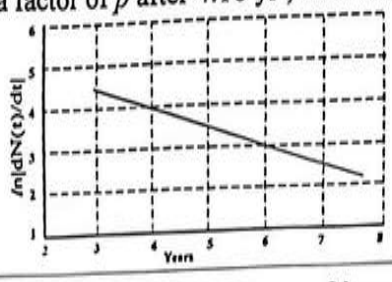
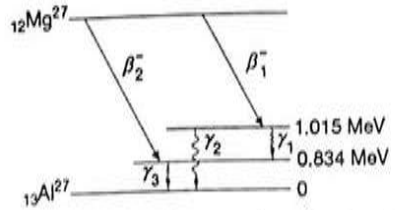
Department of Mechanical Engineering

Date	29 January 2025	Maximum Marks	Quiz 10 + Test 50
Course Code	21ME751Q	Duration	120 Min
Course Name	Nuclear Engineering	USN:	

Q	PART-A (Quiz)	M
1.1	Calculate the binding energy of an alpha particle if mass of $^{1}_{1}H$ atom is 1.007826 u, mass of neutron is 1.008665 u, mass of $^{4}_{2}He$ atom is 4.00260 u. Take 1 u = 931 $MeVc^{-2}$.	1
1.2	Find the increase in mass of water when 1 kg of water absorbs 4.2×10^{3} J of energy to produce a temperature rise of 1 K.	1
1.3	The binding energy of $^{35}_{17}Cl$ nucleus is 298 MeV. Calculate its approximate atomic mass. Given that, mass of proton is m_p = 1.007825 amu and mass of neutron is m_n = 1.008665 amu.	1
1.4	The electric potential energy (in **MeV**) of interaction due to the electric repulsion between two nuclei of ^{12}C when they touch each other at the surface is given by..... Assume that the potential energy of interaction between two nuclei is given by $U = \frac{1}{4\pi\varepsilon_0}\frac{Q_1Q_2}{r}$.	1
1.5	A radioactive nucleus X decays to a stable nucleus Y. Then the graph of rate of formation of Y against time t will be given as	1
1.6	A nucleus X initially at rest, undergoes alpha decay according to the equation $^{232}_{Z}X \rightarrow {}^{A}_{90}Y + \alpha$. What fraction of the total energy released in the decay will be the kinetic energy of the alpha particle?	1
1.7	At time t = 0, N_1 nuclei of decay constant λ_1 and N_2 nuclei of decay constant λ_2 is mixed. The decay rate of mixture is	1
1.8	Sun radiates energy in all direction. The average energy received at earth is 1.4 kWm^{-2}. The average distance between the earth and the sun is 1.5×10^{11} m. If this energy is released by conservation of mass into energy, then the mass lost per day by the sun is approximately	1
1.9	Nuclei of radioactive element A are produced at a rate t^2 at any time t. The element A has decay constant λ. Let N be the number of nuclei of element A at any time t. At time t = t_0, dN/dt is minimum. Then the number of nuclei of element A at time t = t_0 is	1
1.10	In a radioactive decay, let N represent the number of residual active nuclei, D the number of daughter nuclei, and R the rate of decay at any time t. Three curves are shown in figure. The correct ones are 	1

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Department of Mechanical Engineering

Q	PART-B (Test)	M
1	In an nuclear lab an experiment is conducted and following result is observed: A radionuclide with half life 1620 sec is produced at a constant rate 1000 nuclei per second. During each decay energy 200 MeV is released. If production of radio nuclides started at t = 0, calculate (i) the rate of release of energy at 3240 second and (ii) the total energy released upto 405 second.	10
2	During an EL presentation on nuclear engineering topic at RVCE to determine the half-life of a radioactive nuclide a group of students shown the graph of $\ln\left\|\frac{dN(t)}{dt}\right\|$ versus t as shown below. It was mentioned that $\ln\left\|\frac{dN(t)}{dt}\right\|$ represents the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 yr , then what was the calculated value of p? 	10
3	For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t. Two radioactive source P (mean life, τ) and Q (mean life, 2τ) have the same activity at $t = 0$. Their rate of change of activities at $t = 2\tau$ are R_P and R_Q, respectively. If $\frac{R_P}{R_Q} = \frac{z}{e}$, calculate the value of z.	10
4	The β-particles and γ-decays happens as shown in the figure below:  Find the kinetic energy of β-particles and radiation frequencies of γ-decay. Given that mass of $^{27}_{12}Mg$ atom is 26.991425 u and mass of $^{27}_{13}Al$ atom is 26.990080 u.	10
5	The nucleus X is unstable was initially at rest and after decay of an α-particle becomes stable Y nucleus as shown: $^{228}_{92}X \rightarrow {}^{A}_{90}Y + \alpha$. At $t = 0$, the emitted a particle enters in a region of space where a uniform magnetic field $\vec{B} = B_o\hat{i}$ and electric field $\vec{E} = E_o\hat{i}$ exist. The α-particle enters in the region with velocity $\vec{V} = V_o\hat{j}$ from $x = 0$. At time $t = (\sqrt{3} \times 10^7)\left\{\frac{m_\alpha}{q_\alpha E_o}\right\}$ seconds, the particle was observed to have speed twice the initial speed v_0, then find (a) the velocity of a particle at time t. (b) the initial velocity v_0 of the α-particle, (c) the binding energy per nucleon of α-particle. Given that: $m(Y) = 221.03$ u, $m(\alpha) = 4.003$ u, $m(n) = 1.009$ u, $m(p) = 1.008$ u, $m_\alpha = (2/3)\times 10^{-26}$ kg, $q_\alpha = 3.2 \times 10^{-19}$ C and 1 u = 931 MeVc^{-2}	10

All the best!

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF CHEMICAL ENGINEERING

Date	**29th Jan 2025**	**Maximum Marks**	**60**
Course Code	**21CH751C**	**Duration**	**120 Min**
Sem	**VII Semester**	**CIE 2**	

SUSTAINABILITY AND LIFE CYCLE ANALYSIS (Institutional Elective)

Note: 1. Answer all the Questions.

		Part A	M	BT	CO
1	a	Name the frame works used for the identification and structuring of environment indicators.	1	1	2
	b	Name the technique that can detect most of the elements of periodic table.	1	1	2
	c	Identify the component of a hair dryer that has the highest environmental impact.	1	3	3
	d	Define traditional engineering.	1	1	3
	e	List two contradictions of LCA.	1	1	2
	f	State Sherwood relationship.	1	1	2
	g	Name one each government data base and commercial data base available for LCA	2	1	1
	h	Analyze and calculate the ratio of the abundance of materials in the Earth's crust (tons) to the annual use of the materials (tons/yr). Assume an approximate for the crust of 2 x 10^{19} metric tons. This mass is based on a 40 km thickness for continental crust and a 3 km thickness for oceanic crust, with an average density of 3 g/cm^3 (3 tons/m^3). The availability of P in abundance is 0.1% and its usage of Tons/Year is 1.53 x10^6.	2	4	2
		Part B			
		List the components of LCA and explain the LCA process in detail.	10	2	3
2	a	With a neat diagram explain Inductively coupled plasma atomic emission spectroscopy ICP AES	07	2	4
	b	List the steps of toxicity characteristic leaching procedure.	03	1	2
3	a	With a neat diagram explain the sustainable development and design thinking.	07	2	3
	b	Explain the three-pronged approach to characterize the foot print	03	2	2
4		Construct an explanation of the chemistry of dry gasification.	10	3	3
5		Analyze the parameters of sustainability and explain why stone, leather, metals, and timber are regarded as sustainable materials.	10	4	4

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Department of Computer Science and Engineering

Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Date	Jan 2025	Maximum Marks	10+50
Course Code	21CS75IE	Duration	120 Min
Sem	07	CIE-II	

PROMPT ENGINEERING

(INSTITUTIONAL ELECTIVES – II (Group G)))

Q. No.	Questions	M	BT	CO
	QUIZ			
	1.1 Give one example where few-shot learning can be used in LLM applications	01	1	1
	1.2 Name one external tool commonly used with LLMs for real-time data retrieval.	01	2	2
	1.3 Which API would be useful for integrating weather data into an LLM application	01	2	2
	1.4 What is the benefit of using an external database for LLM-based question-answering tasks ?	01	1	1
	1.5 What type of external data would improve the performance of an LLM used for financial predictions ?	01	1	1
	1.6 Which factor is crucial for the successful deployment of prompts in production environments ?	01	1	1
	1.7 Give one example of a task where information extraction from social media could be useful?	01	1	2
	1.8 What does "NER" stand for in the context of information extraction?	01	2	1
	1.9 Provide an example of an entity that could be extracted from the sentence "Schedule a meeting with John tomorrow at 3 PM	01	1	1
	1.10 In the context of API integration, what does REST stand for ?	01	1	1
	TEST			
2.	Summarization using sources is an essential task for LLMs. How can LLMs leverage external sources to generate more precise and reliable summaries? Discuss the methodology for retrieving relevant information from external data sources, synthesizing it, and generating a coherent summary. Provide examples from	10	1	1

Academic year 2024-2025 (Odd Sem)

3.	Explain the best practices in prompt engineering for large language models (LLMs). Discuss how prompt design influences model performance and how to craft clear, specific, and concise prompts for effective results. Provide examples of both effective and ineffective prompts.	10	1	2
4.	Advanced prompting techniques often involve the combination of multiple approaches. Discuss how advanced prompting methods like few-shot learning, zero-shot learning, and chain-of-thought prompting are applied in real-world scenarios. Compare their effectiveness and challenges in different use cases. task without any explicit examples, relying solely on the clarity of the instructions.	10	4	5
5.	Explain the role of information extraction in function calling. How can a conversational agent extract specific information (such as dates, times, and names) from user input, and how does it invoke functions to act on this extracted data?	10	3	4
6.	Explain the different techniques used in document parsing for information extraction. How can complex documents such as contracts, financial statements, and academic papers be parsed to extract meaningful information for automation and decision-making processes?	10	2	3

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BT-Blooms Taxonomy, CO-Course Outcomes

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test & Quiz	Max Marks	17	13	10	10	10	27	13	10	-	-

urse Outcomes:

:	Demonstrate an understanding of prompt engineering principles including how prompt structure and phrasing impact the performance of AI models.
	Design and implement effective prompts to create and apply prompts for various natural language processing (NLP) tasks, such as text generation, summarization, and translation, using AI models..
	Critically evaluate the effectiveness of prompts - assess the quality and performance of prompts in terms of accuracy, coherence, and relevance, identifying
	Apply prompt engineering

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	29/01/2025	Maximum Marks	10+50
Course Code	21HS75IR	Duration	120 Mins
Sem	VII	CIE – II	

PRINCIPLES & PRACTICES OF CYBER LAW

Note:

1. Answer all the Questions.

Q.No.	PART- A	M	BT	CO
1	List any two types of E-commerce models.	2	2	CO2
2	What are 'Digital Signature' and its use in E-commerce?	2	2	CO2
3	What is the main concern regarding copyright infringement in cyberspace?	2	2	CO3
4	How does cybersquatting affect trademark owners?	2	2	CO3
5	How can a trademark owner protect their brand online?	2	2	CO4
	PART -B			
1	A company named XYZ Ltd. enters into a contract with ABC Ltd. through an online platform. The contract is electronically signed by representatives of both parties using Digital Signatures issued by a recognized Certifying Authority. However, later, XYZ Ltd. disputes the validity of the contract, claiming that one of the signatories had not authorized the use of their digital signature. (i)What role does a Digital Signature play in validating an electronic contract? (ii)What legal provisions under the IT Act 2000 support the enforceability of this electronic contract?	10	4	CO2
2	A company launches a new product and discovers a similar domain name being used to divert customers to a competitor's site. Propose a comprehensive strategy to address this issue legally and technologically with one case example.	10	4	CO2
3	Bring out the difference between Digital Signature & Electronic Signature along with the advantages and disadvantages.	10	2	CO3
4	Design a comprehensive strategy for an e-commerce business to protect customer transactions using digital signatures and cryptography.	10	3	CO4
5a	If an individual finds that their copyrighted work has been uploaded to a website without their consent, what steps should they take to protect their rights.	05	2	CO4
5b	Analyze the impact of piracy on digital content creators in cyberspace.	05	2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	30				20	10	20		

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	28.01.2025	Maximum Marks	05 + 25
Course Code	21IM78	Duration	60 Mins
Sem	VII Semester	CIE – II	

INNOVATION AND ENTREPRENEURSHIP

Note:
1. Answer all the Questions.
2. Use suitable sketches and tabular column wherever necessary.
3. Observe the marks split-up and answer accordingly.

PART – A

Sl. No.	Questions	M	BT	CO
1	List any two rules of brainstorming.	01	L1	CO1
2	Imagine you are starting your ice-cream outlet. List 2 ways in which you can make sure your funds are used optimally.	01	L2	CO2
3	Identify two bad practices students are doing to get "AICTE activity points". Suggest two methods to curtail the same.	02	L4	CO2
4	Comment on the importance of "Early Adopters".	01	L3	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Discuss the four types of Markets (Use tabular column format) and comment on the most suitable market type for a start-up venture.	05	L2	CO2
2.	When you are allowed to form teams on your own, lack of coordination among the team members is clearly visible and reflects in the outcome of the work also. As individuals, each of the students is competent in their own way. Use design thinking principles to identify the problem with students and how it can be resolved. (Concentration should be on the first three stages of DT)	05	L4	CO3
3.	Illustrate the lean canvas of business model for your practice venture using Ash Maurya's template and assess each of the 9 blocks using a 5-point scale (1 being lowest and 5 being highest). Identify any 3 blocks which needs improvements and suggest strategies to improve the same.	05	L3	CO4
4.	Construct 2 detailed personas of students belonging engineering education from your college and propose 2 problem definitions using DT framework.	07	L5	CO3
5.	Choose any 1 problem definition from the previous question and answer the following. i. Assess & depict the input requirements based on your understanding the solution space. (2 marks) ii. Sketch the solution. (1 mark)	03	L5	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks	Max	Quiz	01	03	01	--	--	01	01	01	02	--	--
		Test	--	05	12	08	--	--	05	05	05	10	--

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